

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2024

Bachelor in Computer Applications

Course Title: **Operating Systems**

Code No: **CACS251** Semester: **IV**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Define kernel. Explain different types of kernel in operating system.
3. What is PCB? What are different information that are included in a PCB?
4. How paging can be used for memory management? Also differentiate it with segmentation.
5. What are page faults and thrashing? Explain demand paging with protection bit.
6. How deadlock can be prevented? Explain in detail.
7. Explain different file access methods in detail.
8. What is distributed system? Explain the role of clock synchronization in distributed system.

Group C

Attempt any TWO questions[2x10=20]

9. What is critical section problem? How semaphores can be used to solve critical section problem Also explain producer consumer problem with its solution using Mutex.
10. What is Belady's Anomaly in page replacement Algorithm? Explain with example. Consider the following page reference string: 7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0, 1. How many page faults will occur if the program has four page frames available to it and use the following page replacement algorithm? a) FIFO replacement b) LRU replacement c) Optimal replacement
11. What is use of DMA controller? Consider the following process and answer the following questions. Process Allocation Max Available [A,B,C,D] [A,B,C,D] [A,B,C,D] P0 0 1 10 0 1 2 1 5 2 0 P1 0 0 0 1 7 5 0 P2 1 3 5 4 2 3 5 6 P3 0 6 3 20 6 5 2 P4 0 0 1 40 6 5 6 a) What is the content of need matrix? b) Is the system in safe state? c) If P2 request (3 4 0 2), can the request be granted immediately.